

Healthcare Analytics: MIMIC-3 on Azure Databricks

Project Objective

- Build an end-to-end scalable data pipeline on Azure & Databricks
- Perform data cleaning and transformation
- Analyze data using EDA techniques
- Develop predictive machine learning models
- Visualize insights for decision-making
- Enable GenAI-based intelligent querying

Tools & Technologies

Cloud & Storage

- Microsoft Azure
- Azure Blob Storage / Data Lake

Data Processing

- Azure Databricks
- Apache Spark (PySpark)

Data Architecture

- Medallion Architecture (Bronze, Silver, Gold)

Programming & Querying

- Python
- Pyspark
- SQL

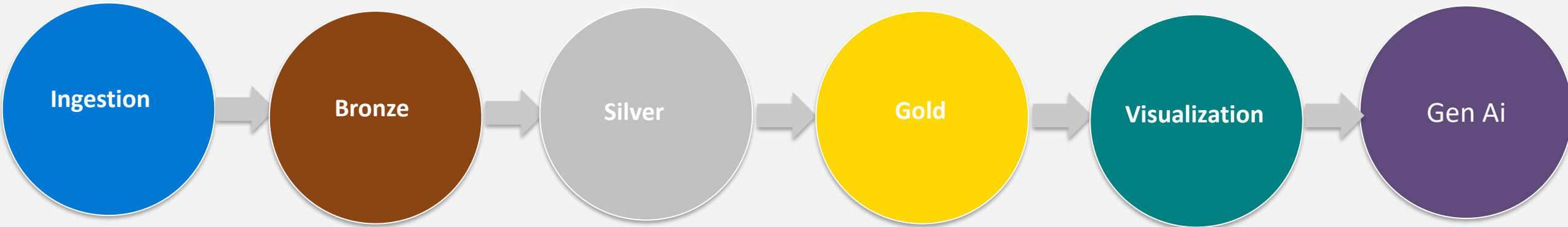
Data Visualization

- Power BI

AI Layer

- GenAI Integration

Project Workflow



Core Components

- 1. **Ingestion:** Azure Blob Storage (Landing Zone for raw data)
- 2. **Medallion Architecture:** Bronze (Raw) → Silver (Cleaned) → Gold (Curated Aggregates)
- 3. **Databricks Processing:** Data cleaning, transformation, and feature engineering
- 4. **Exploratory Data Analysis (EDA):** Identify patterns, distributions, and trends
- 5. **Predictive Modeling:** Mortality prediction using Random Forest and Gradient Boosting
- 6. **Data Visualization:** Interactive charts for insights on mortality, age, diagnosis, LOS
- 7. **GenAI Integration:** AI-powered clinical insights using curated Delta Lake data
- 8. **Security & Governance:** Secure trial accounts and data access control in Azure environment

Azure Resource Setup

- Created an Azure Resource Group (*rg-databricks*) to manage project resources
- Deployed Azure Data Lake Storage and Azure Databricks workspace (*dataproject123*) for data processing

Home > Resource Manager | Resource groups

 **rg-dataengineering**
Resource group

 Export resource groups using Bicep or Terraform

Help me generate Terraform for this resource group configuration. 



 Create

 Manage view 

 Delete resource group

 Refresh

 Export to CSV

 Open query

|

 Assign tags



 Group by none 

 Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

> Settings

> Cost Management

> Monitoring

> Automation

> Help

 Essentials 

Resources

Recommendations

 Filter for any field...

Type equals all 

Location equals all 

 Add filter

☐	Name ↑		Type	Location
☐	 datalakeproject123		Storage account	East US
☐	 rg-databricks		Azure Databricks Service	East US

Add or remove favorites by pressing Ctrl+Shift+F

Data Upload to Data Lake

- Uploaded MIMIC-3 CSV files into Azure Data Lake Storage Gen2 using hierarchical storage
- Organized raw healthcare data in the landing zone for scalable and secure processing

The screenshot displays the Azure Data Lake Storage Gen2 interface for a container named 'mimic-data'. The interface includes a left-hand navigation pane with options like 'Overview', 'Diagnose and solve problems', 'Access Control (IAM)', and 'Settings'. The main area shows a list of 25 items, which are CSV files. The files are organized in a table with columns for Name, Last modified, Access tier, Blob type, Size, and Lease state. The files are all 'Block blob' type and 'Available' state. The files listed are: ADMISSIONS.csv, CALLOUT.csv, CAREGIVERS.csv, CHARTEVENTS.csv, CPTEVENTS.csv, DATETIMEEVENTS.csv, DIAGNOSES_ICD.csv, DRGCODES.csv, D_CPT.csv, and D_ICD_DIAGNOSES.csv. The interface also features a search bar, a filter button, and a dropdown menu to 'Only show active blobs'.

Name	Last modified	Access tier	Blob type	Size	Lease state
ADMISSIONS.csv	4/6/2026, 10:46:10 PM	Hot (Inferred)	Block blob	26.19 KiB	Available
CALLOUT.csv	4/6/2026, 10:53:36 PM	Hot (Inferred)	Block blob	13.5 KiB	Available
CAREGIVERS.csv	4/6/2026, 10:46:29 PM	Hot (Inferred)	Block blob	173.97 KiB	Available
CHARTEVENTS.csv	4/6/2026, 10:59:36 PM	Hot (Inferred)	Block blob	74.13 MiB	Available
CPTEVENTS.csv	4/6/2026, 10:53:06 PM	Hot (Inferred)	Block blob	145.53 KiB	Available
DATETIMEEVENTS.csv	4/6/2026, 10:56:54 PM	Hot (Inferred)	Block blob	1.7 MiB	Available
DIAGNOSES_ICD.csv	4/6/2026, 10:46:47 PM	Hot (Inferred)	Block blob	47.85 KiB	Available
DRGCODES.csv	4/6/2026, 10:57:36 PM	Hot (Inferred)	Block blob	22.58 KiB	Available
D_CPT.csv	4/6/2026, 11:12:12 PM	Hot (Inferred)	Block blob	12.42 KiB	Available
D_ICD_DIAGNOSES.csv	4/6/2026, 11:01:58 PM	Hot (Inferred)	Block blob	1.25 MiB	Available

Azure Databricks Workspace

- Launched Azure Databricks workspace for data engineering and analytics
- Used notebooks to process data, perform EDA, and build machine learning models

Home > rg-dataengineering



List access control settings for this Azure Databricks Service.

How do I troubleshoot issues with this Azure Databricks Service?

Summarize the jobs running on this Azure Databricks Service.



Search

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

> Settings

> Automation

> Help

Essentials

JSON View

Status : Active

Workspace type : Hybrid

Resource group : [rg-dataengineering](#)

Managed Resource Group : [databricks-rg-rg-databricks-kmoku6z5fynm6](#)

Location : East US

URL : <https://adb-7405610116102176.16.azuredatabricks.net>

Subscription : [Azure subscription 1](#)

Pricing Tier : [Standard \(Apache Spark, Secure with Microsoft Entra ID\)](#) (Click to ch...

Subscription ID : 17c744be-6f4b-4c42-a79e-06e897a34917

Enable No Public IP : Yes

Tags ([edit](#)) : [Add tags](#)



Launch Workspace

Upgrade to Premium

Medallion Architecture

1. Bronze Layer (Raw Data)

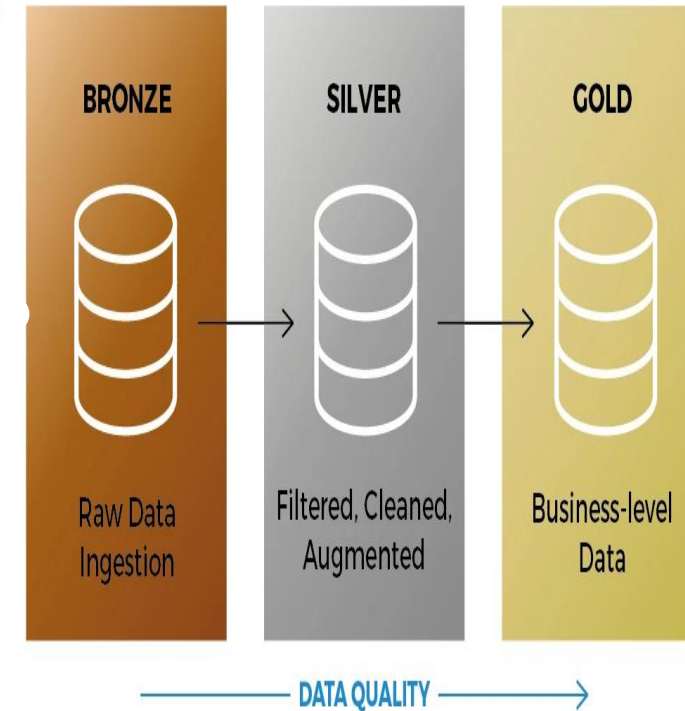
- Stores ingested data exactly as received
- Contains all original records without modifications
- Acts as the source for downstream processing

2. Silver Layer (Cleaned Data)

- Handles missing or corrupt data
- Standardizes formats and applies basic transformations
- Enriches data with feature engineering for analytics

3. Gold Layer (Curated Data)

- Aggregated and refined for business use
- Optimized for reporting and machine learning models
- Supports fast, reliable queries for decision-making



Predictive Modeling on Integrated Healthcare Data

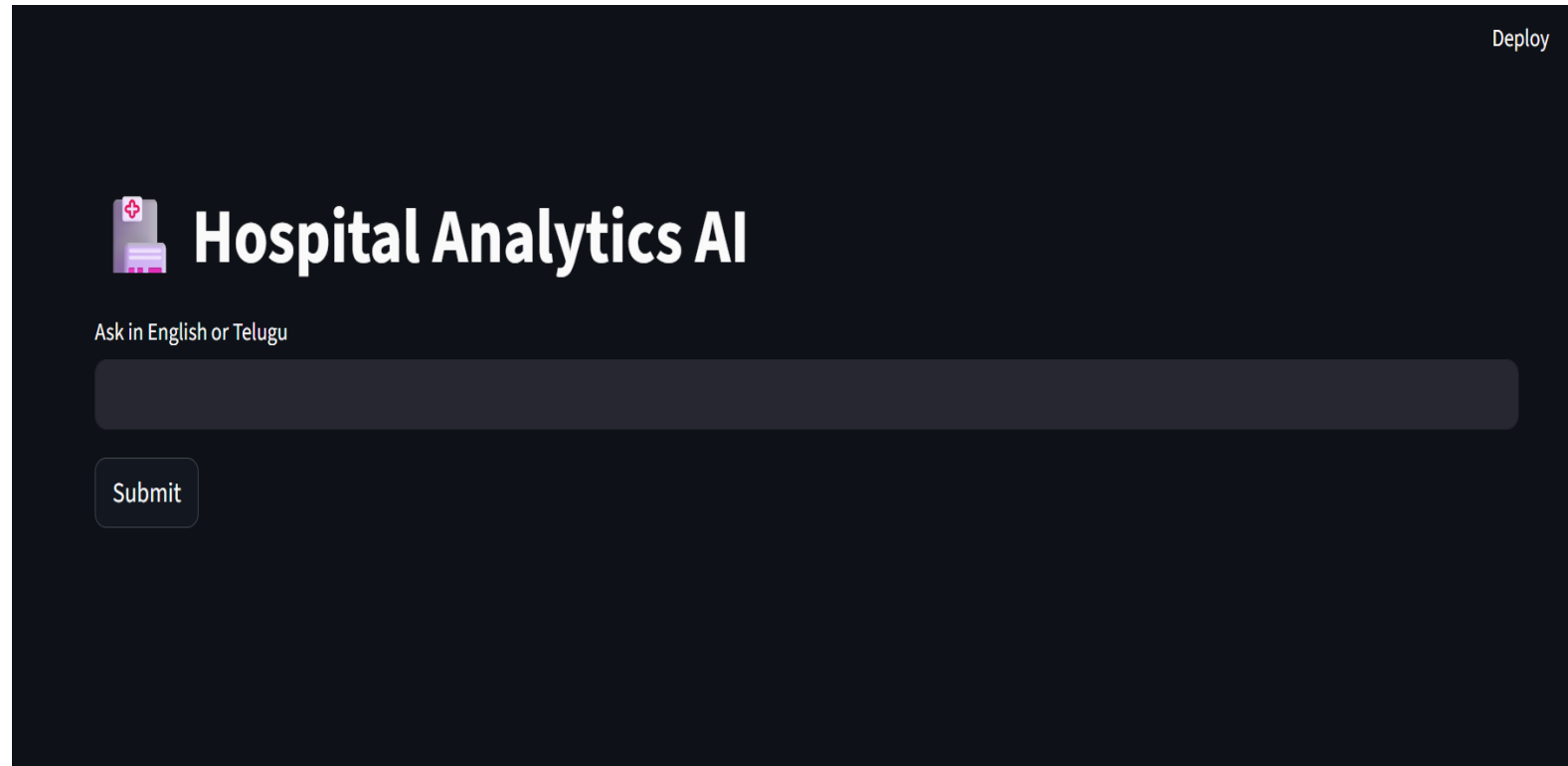
- Combined and processed datasets through Silver and Gold layers for consistency
- Applied supervised machine learning classification techniques on the unified dataset
- Used linear, tree-based, and distance-based models to capture different data patterns
- Performed hyperparameter tuning to optimize model performance and improve generalization
- Tuned parameters like tree depth, number of estimators, and neighbor values
- Split data into training and testing sets for reliable evaluation
- Evaluated models using Accuracy and ROC-AUC metrics
- Achieved high model performance (>0.85 accuracy), with tree-based models performing best
- Identified key predictors such as age, diagnosis type, and hospital stay duration
- Enabled early identification of high-risk patients using predictive insights

Data Visualization & Dashboards

- Developed interactive dashboards using **Power BI** to visualize integrated healthcare data
- Created visual reports for trend analysis, outcome comparison, and pattern identification
- Designed dashboards to present insights in a clear and user-friendly manner
- Enabled dynamic filtering and drill-down capabilities for deeper data exploration
- Supported data-driven decision-making through intuitive and real-time visual insights

GenAI Integration

- Collected and processed multiple CSV files, storing the unified data in **MySQL tables** for structured access
- Enabled intelligent querying of the database using **GenAI-powered Large Language Models (LLMs)**
- Automatically generated **SQL queries** based on natural language questions to fetch precise answers from the tables
- Delivered real-time, accurate insights by converting user queries into actionable outputs, enhancing data accessibility and decision-making



Key Insights from the Project

- Successfully ingested and integrated multi-source healthcare data into a unified, analytics-ready dataset
- Medallion architecture ensured **data quality, consistency, and layered processing** for robust analytics
- Machine learning models achieved **high predictive performance**, enabling early identification of high-risk patients
- Power BI dashboards provided **interactive visual insights** for trends, distributions, and outcomes
- GenAI integration allowed **intelligent querying** of structured data with automatic SQL generation for fast insights

*Thank
you!*